ALICE L WALTON SCHOOL OF MEDICINE

Antihyperlipidemic Drugs– Statins

Last updated November 4, 2025 · 12 min read · ScholarRx

Adapted by: Diptiman Bose, Trager Hintze

Editors: Jordan Finn, Russell Johnson

Adapted from: Antihyperlipidemic Drugs

Listen to this Brick

0:00 / 21:13 1x

Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Explain the mechanism of action of statins in the context of managing hyperlipidemia and the rationale behind their use in preventing cardiovascular diseases.
2. Classify statins based on their dose and discuss their Pharmacokinetic/pharmacodynamic profiles.
3. Explain the recautions associated with statins and discuss strategies to minimize/prevent such effects.
4. Explain the common reasons for statin intolerance and provide alternative therapeutic strategies.
5. Identify the potential drug-drug interactions of statins and explain their underlying mechanisms.

CASE CONNECTION

"Those ads for cholesterol medications are getting to me, and now I'm wondering about my cholesterol." This is how JC, a 45-year-old male, begins the conversation with you during his appointment. JC admits he doesn't exercise and continues to smoke about three-quarters of a pack of cigarettes every day. His physical exam is unremarkable except for a body mass index of 27 kg/m^2. You decide he should have a lipid profile. His results are abnormal: his low-density lipoprotein (LDL) cholesterol is 200 mg/dL, high-density lipoprotein (HDL) cholesterol is 32 mg/dL, and triglycerides are 125 mg/dL. You talk to him about healthy lifestyle choices, and he is prescribed atorvastatin.

Thinking back to JC, how do you explain his new drug? What advice would you give JC about this medication? Consider your answers as you read, and we'll revisit JC at the end of the brick.

GO TO CONCLUSION ↓

What Are Statins? (LO#1)

Statins are pharmacologically classified as antihyperlipidemic agents. Antihyperlipidemic agents lower blood levels of lipids, including low-density lipoprotein (LDL) and cholesterol. They are used to treat lipid disorders (dyslipidemia), which often present with high levels of cholesterol and/or triglycerides (TGs). These disorders are most often seen in otherwise well patients but can also be secondary to conditions like diabetes mellitus or renal or liver disease or more rarely from genetic abnormalities of lipid metabolism.

What is Hyperlipidemia?

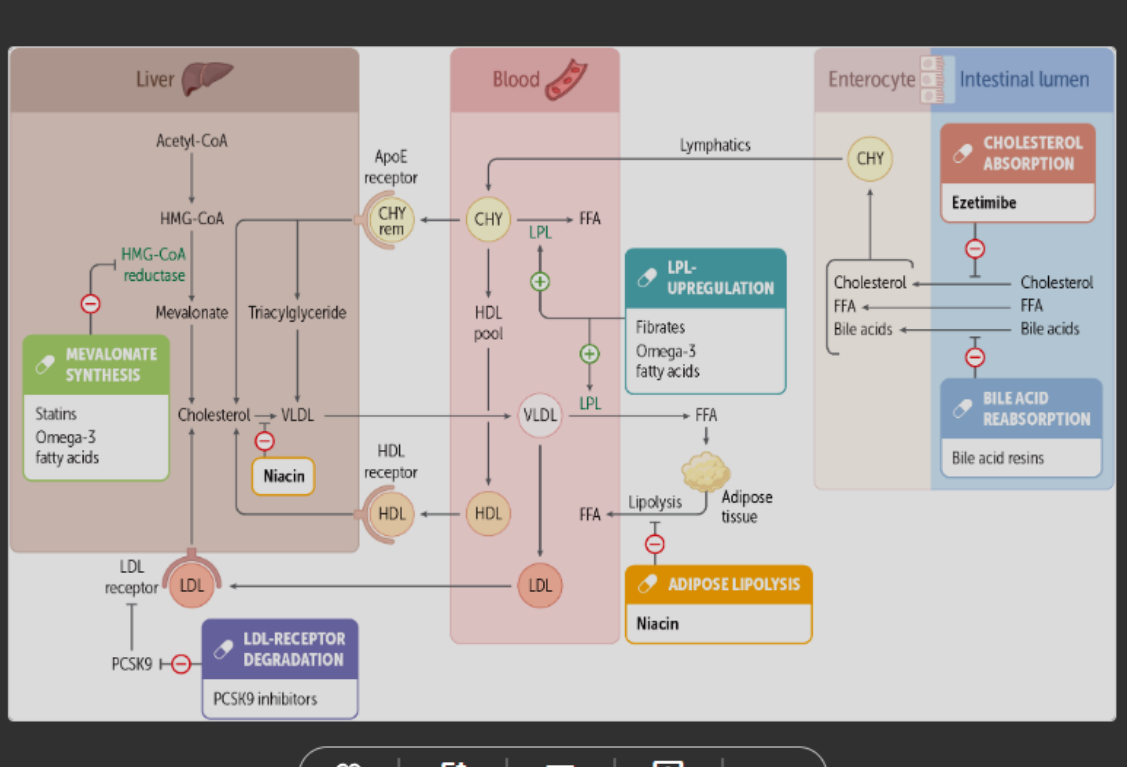
Hyperlipidemia is usually asymptomatic and commonly detected when performing screening tests during a routine checkup. However, over time, hyperlipidemia can lead to atherosclerosis, which increases the risk of cardiovascular disease, leading to chest pain, myocardial infarction (MI), and strokes, among other complications. Hyperlipidemia can cause morbidity or mortality in a majority of middle-aged adults and has the cause of one-third of all deaths in this age range. Dyslipidemia, which includes hyperlipidemia, and low levels of HDL-C increase atherogenic risk. Genetic disorders and lifestyle factors can contribute to dyslipidemia.

What pharmacological agents are used to treat hyperlipidemia?

The treatment goals for hyperlipidemia should be patient-focused; however, most are designed to lower LDL-C and non-HDL-C lipoproteins. Statins are one the most commonly prescribed, an effective and well-tolerated drug class used for dyslipidemia. There are 6 different statins approved by the US-FDA and they all end in "statin"!! Examples of statins are Atorvastatin, Fluvastatin, Lovastatin, Pitavastatin, Pravastatin, Rosuvastatin, and Simvastatin.

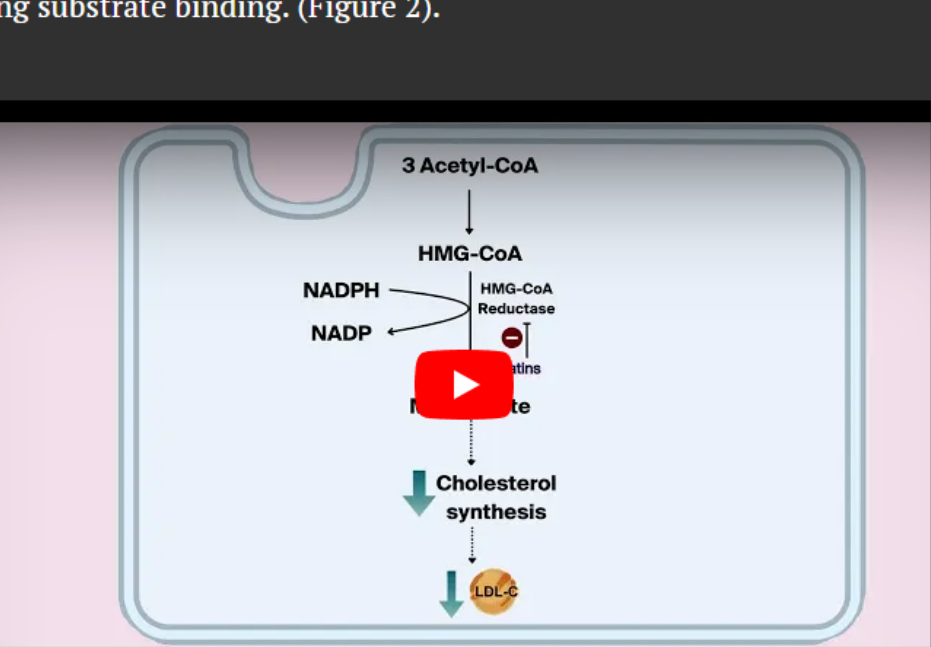
How are lipoproteins circulating?

In your BOOM course, we have discussed the lipid metabolism and the synthesis of lipoproteins. Figure 1 reviews lipid metabolism and provides an overview of how the various lipoproteins are synthesized, stored, and regulated by the body. Before we discuss the statin pharmacology, it will be good to review the process (Figure 1).

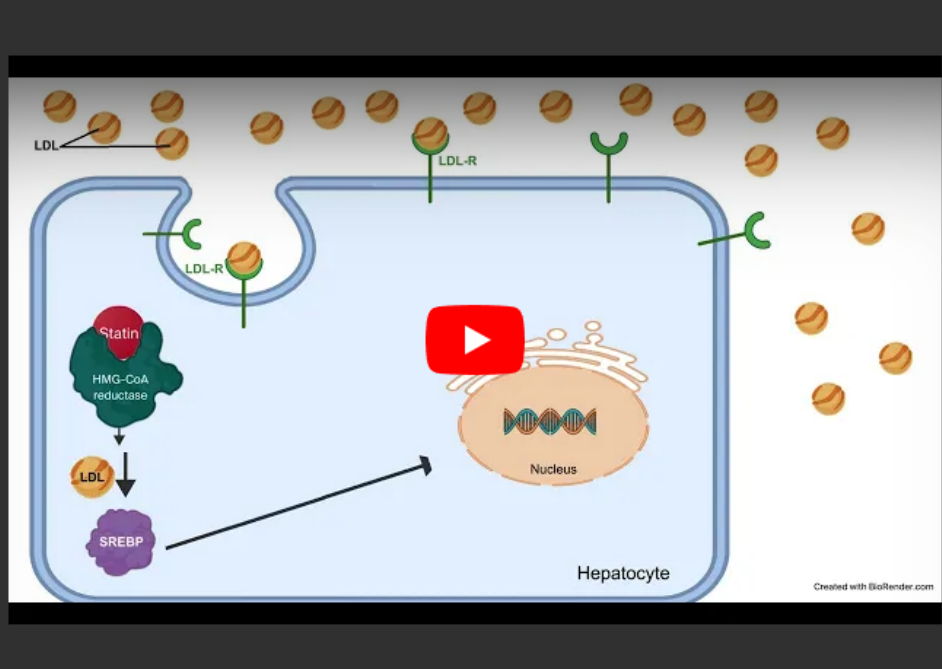


HOW DO STATINS WORK?

Statins inhibit the 3-hydroxy-3-methylglutaryl-coenzyme A (HMG-CoA) reductase, the rate-limiting enzyme in the pathway to convert acetyl-CoA to cholesterol. Statins occupy a portion of the binding site for the HMG-CoA blocking substrate binding. (Figure 2).



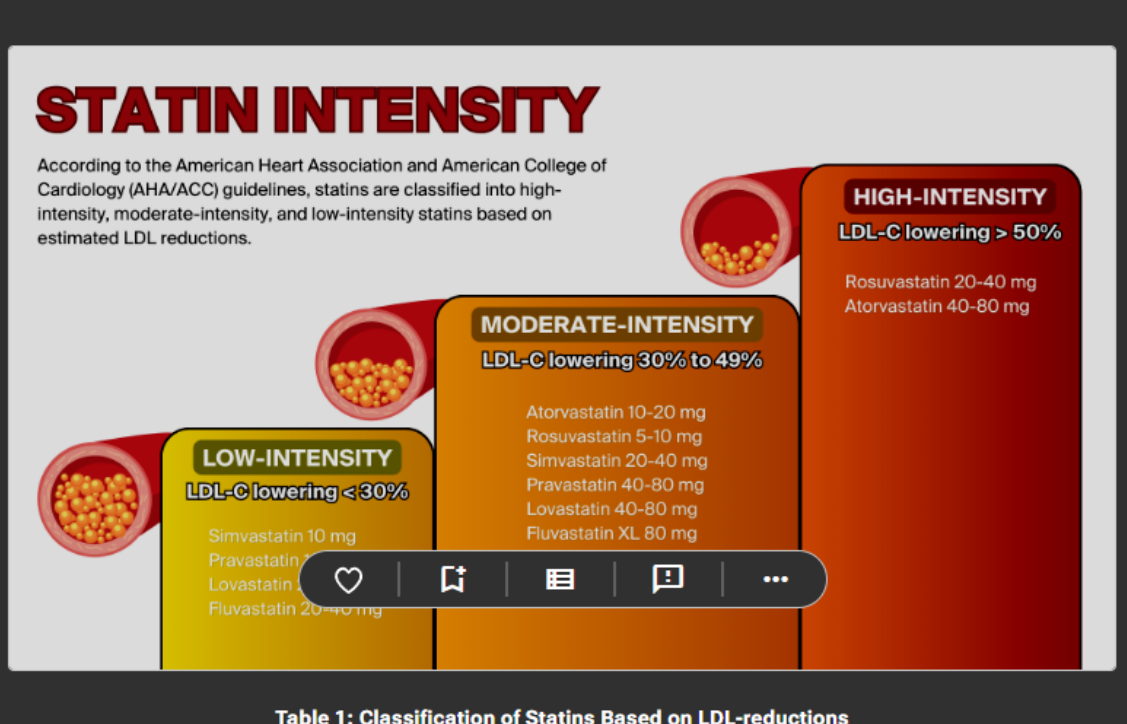
- By inhibiting the HMG-CoA activity, statins can reduce circulating LDL levels, thus decreasing the production of Cholesterol.
- In response to the decrease in circulating LDL, the expression of plasma membrane LDL-P es LDL-R receptor expression is shown in the figure below. An increase in LDL-R results in the removal of LDL from the blood, thus reducing LDL-C levels. (Figure 3)
- Most statins modestly increase HDL levels (~5%)
- A decrease in circulating VLDL levels help decrease TG levels (~20-40%)



- **Lipid-lowering** has been proven to improve survival in patients with hypercholesterolemia for primary prevention (no clinical evidence of coronary disease) and secondary prevention (with established coronary disease). This benefit has been shown when cholesterol levels are "normal" or even borderline "high." While lowering lipid levels can explain these benefits, other factors also play a role.
- **Regression of artherosclerosis:** Regression of artherosclerotic lesions occurs with statin thickness. Exact timing is not clear, but some studies have reported occurrence within 12-24 months of therapy
- **Plaque stabilization:** Coronary plaque rupture can lead to coronary syndromes. Lipid lowering can contribute to maintaining the integrity of the fibrous cap of the plaque, thus protecting rupture.
- **Reversal of endothelial dysfunction:** The vasoconstriction associated with endothelial dysfunction is reduced with statin therapy, however the mechanism underlining this effect is not clear.

How do we classify statins? (LO#2)

According to the American Heart Association and American College of Cardiology (AHA/ACC) guidelines, statins are classified into high-intensity, moderate-intensity, and low-intensity statins based on estimated LDL reductions (Table 1).



STATIN INTENSITY

According to the American Heart Association and American College of Cardiology (AHA/ACC) guidelines, statins are classified into high-intensity, moderate-intensity, and low-intensity statins based on estimated LDL reductions.

HIGH-INTENSITY
LDL-C lowering > 50%
Rosuvastatin 20-40 mg
Atorvastatin 40-80 mg

MODERATE-INTENSITY
LDL-C lowering 30% to 49%
Atorvastatin 10-20 mg
Rosuvastatin 5-10 mg
Simvastatin 20-40 mg
Pravastatin 40-80 mg
Lovastatin 40-80 mg
Fluvastatin XL 80 mg

LOW-INTENSITY
LDL-C lowering < 30%
Simvastatin 10 mg
Pravastatin
Lovastatin
Fluvastatin 20-40 mg

Table 1: Classification of Statins Based on LDL-reductions

How do statins reduce LDL levels?

WHAT ARE ADVERSE EFFECTS, CONTRAINDICATIONS, AND PRECAUTIONS WITH STATIN USE? (LO#3 and 4)

- ***Hepatic dysfunction***: All statins are taken orally and have low systemic bioavailability since they undergo extensive first-pass metabolism by the liver. Thus, monitoring liver function is important with statin use. Statins can elevate liver aminotransferases in patients receiving statins within the first three months of therapy and are dose-dependent. A liver function test (LFT) is usually ordered before initiating statin therapy and is repeated only after clinica using alternate agents is recommended in patients with three times upper ALT levels with a confirmed second lab test.
- ***Stain-associated muscle symptoms (SAMS):*** SAMS is one of the most common adverse effects of statin use. SAMS can present as myalgia (muscle aches, soreness, tenderness, or cramps), myopathy (muscle weakness ± CPK elevations), myositis (inflammation), or clinical rhabdomyolysis (significantly elevated CPK levels with myoglobinuria and acute kidney injury). Risk of myopathy is lowest with Fluvastatin, pravastatin, and pitavastatin since they are not CYP3A4 metabolized. On the other hand, SAMS has a higher risk with the use of Lovastatin, Simvastatin, and Atorvastatin. Simvastatin dose is maxed at 40mg/day to prevent SAMS. Hypothyroidism can also predispose patients for SAMS. Low vitamin D levels are associated with high rates of myalgia and myopathy.
- ***GI effects***: Diarrhea is a common adverse effect of the drug class.
- ***Risk in Pregnancy and breastfeeding***: Statins should be discontinued in pregnant patients and breastfeeding is not recommended in patients taking a statin.
- ***Diabetes mellitus***: Statin therapy can cause an increased risk of developing diabetes by affecting glucose metabolism in nondiabetics or interfering with glycemic control in patients with diabetes. Just as statins can increase LDL-R in the liver and increase uptake of LDL-C in the liver, the same occurs in the pancreas. This can increase the chances of lipotoxicity in the β-cells resulting in decreased insulin secretion and loss of glycemic control.

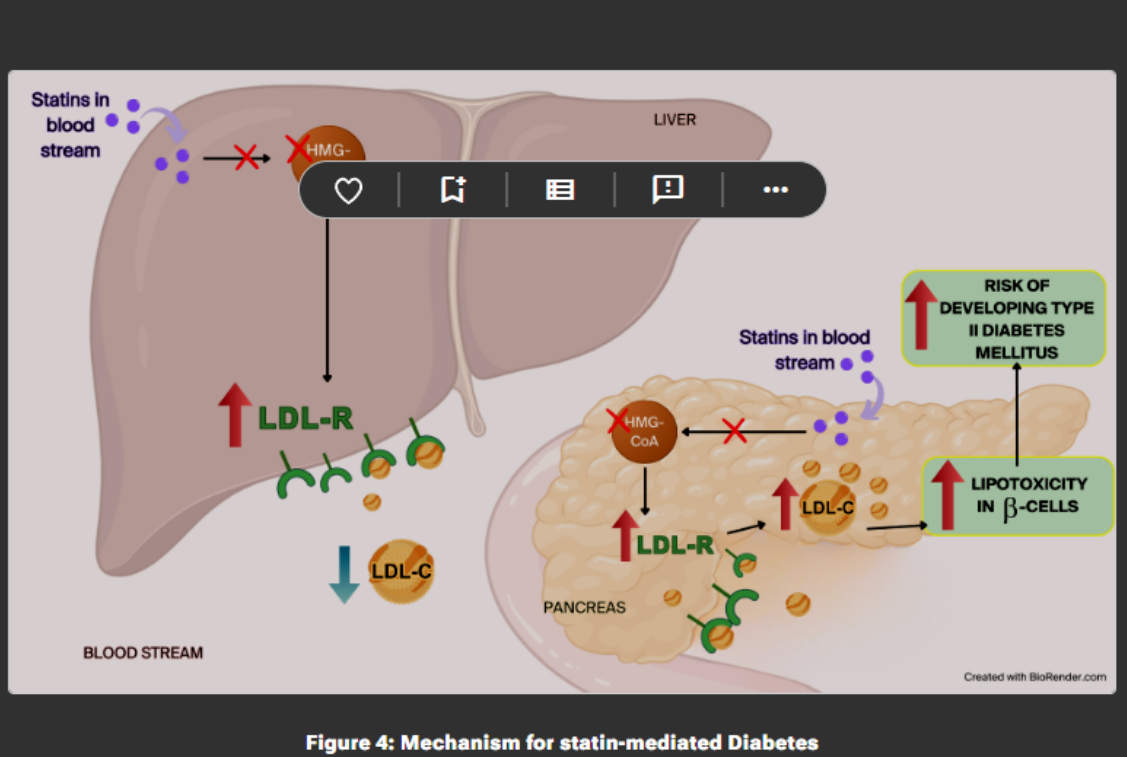


Figure 4: Mechanism for statin-mediated Diabetes

Doses higher than 40 mg of Atorvastatin would be considered ___ intensity statin

What are some potential drug interactions with statin use? (LO#5)

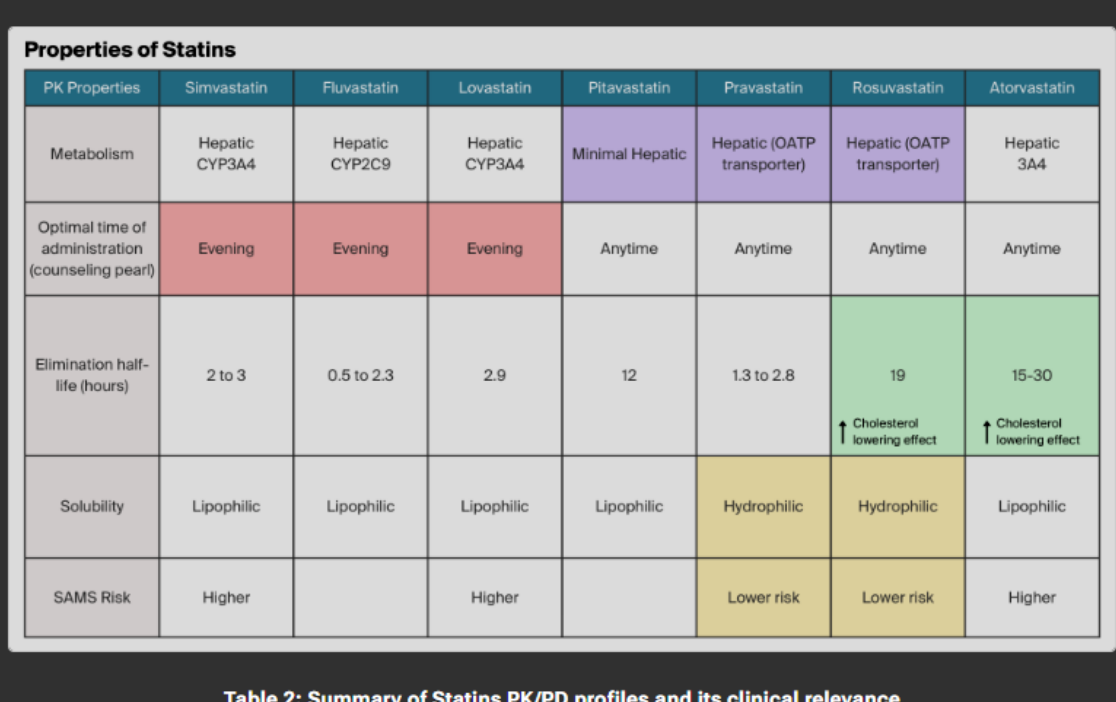
- Simvastatin, lovastatin, and atorvastatin are metabolized by CYP3A4. Fluvastatin clearance is partly dependent upon CYP2C9 metabolism. Rosuvastatin, pitavastatin, and pravastatin are cleared primarily by non-CYP450 transformations. Hence, drugs that inhibit CYP3A4 can increase the risk of statin myopathy when administered with statins extensively metabolized by CYP3A4. Drugs such as glucocorticoids, daptomycin also increase the risk of myopathy and can increase CPK levels with statin use.
- Calcium channel blockers –Calcium channel blockers amlodipine, diltiazem and verapamil are moderate inhibitors of CYP3A4 metabolism and can increase ADRs in statins metabolized by CYP3A4.
- Amiodarone – Amiodarone, an anti-arrhythmic drug, inhibits CYP3A4 metabolism. The risk for myopathy and rhabdomyolysis with simvastatin in doses greater than 20 mg/day increases with concurrent Amiodarone use.
- Grapefruit juice—Grapefruit juice inhibits intestinal CYP3A4, and daily consumption of the juice with Simvastatin increases its level. Rosuvastatin, pravastatin, and pitavastatin is recommended for patients who consume large amounts of grapefruit or grapefruit juice.
- Genetics does play a role in the variability in the response to and side effects of statins. For instance, CYP2D6 phenotype appears to be important in patients treated with simvastatin, as it can affect both the degree of lipid-lowering and tolerability. Part of that may be related to genetic differences in the rate of drug metabolism. As an example, a lower initial dose of rosuvastatin is recommended in Asian individuals since CYP2D6 is functionally absent in 7 percent of White and African American individuals, while deficiency is rare among Asian persons.

Which statins have the lowest risk of myopathy?

What is the relevance of understanding statins PK/PD profiles? (LO#2 and 5)

Table 2 summarizes information that is very important when using statins. The key take-aways are highlighted:

- Metabolism: Pitavastatin has minimal hepatic metabolism, hence can be used in patients with hepatic dysfunction (purple).
- Time of administration: Patients on simvastatin and lovastatin should be advised to take them in the evening to maximize efficacy of the medication.
- Half-life: You do not need to know all the $t_{1/2}$ for each statin but note that Rosuvastatin and Atorvastatin have longer $t_{1/2}$; thus, they are categorized as high intensity statins.
- SAMS risk: Prava risk of SAMS because they are more hydrophilic



Properties of Statins

PK Properties	Simvastatin	Fluvastatin	Lovastatin	Pitavastatin	Pravastatin	Rosuvastatin	Atorvastatin
Metabolism	Hepatic CYP3A4	Hepatic CYP2C9	Hepatic CYP3A4	Minimal Hepatic	Hepatic (OATP transporter)	Hepatic (OATP transporter)	Hepatic 3A4
Optimal time of administration (counseling pearl)	Evening	Evening	Evening	Anytime	Anytime	Anytime	Anytime
Elimination half-life (hours)	2 to 3	0.5 to 2.3	2.9	12	1.3 to 2.8	19 ↑ Cholesterol lowering effect	15-30 ↑ Cholesterol lowering effect
Solubility	Lipophilic	Lipophilic	Lipophilic	Lipophilic	Hydrophilic	Hydrophilic	Lipophilic
SAMS Risk	Higher		Higher		Lower risk	Lower risk	Higher

Table 2: Summary of Statins PK/PD profiles and its clinical relevance

ACTIVE LEARNING QUESTION

Match the statin to its PK/PD property

Simvastatin	✓ Administer in the evening
Rosuvastatin	✓ Hydrophillic
Pitavastatin	✓ Minimal Hepatic metabolism
Atorvastatin	✓ Higher SAMS risk

Explanation

An important clinical pearl is to remind the patient that Simvastatin should be taken in the evening. Since Rosuvastatin is hydrophilic, it has a lower SAMS risk; on the other hand, atorvastatin is lipophilic, and has a higher SAMS risk. Pitavastatin has minimum hepatic metabolism and can be used in case of hepatic insufficiency.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to JC, how do you explain his new drug? What advice would you give JC about this medication?

You explain to JC that the results of his lipid profile put him at an increased risk of heart and vascular disease. "Atorvastatin blocks an early step in the production of cholesterol. Atorvastatin also lasts longer (more half-life) and therefore can have a better impact on reducing cholesterol levels." After this conversation and a review of the adverse effects of the medication, you discuss the possibility of muscle pain. You can emphasize that JC can take this medication any time of the day and avoid drinking grapefruit juice (grapefruit inhibits CYP3A4).

Summary

- Statins are antihyperlipidemic agents that lower blood levels of lipids, including low-density lipoprotein (LDL) and cholesterol, and are used to treat lipid disorders.
- Hyperlipidemia, often leading to atherosclerosis, increases the risk of cardiovascular disease, including chest pain, myocardial infarction, and strokes.
- Statins work by inhibiting the 3-hydroxy-3-methylglutaryl-coenzyme A (HMG-CoA) reductase, thus reducing circulating LDL levels.
- Statins are classified into high-intensity, moderate-intensity, and low-intensity based on estimated LDL reductions.
- Adverse effects of statins include hepatic dysfunction, muscle symptoms, gastrointestinal effects, risk in pregnancy and breastfeeding, and increased risk of developing diabetes.
- Statins can interact with certain drugs and food, and genetic differences can influence the response to and side effects of statins.

Review Questions

1. A 65-year-old female comes to the clinic complaining of new-onset muscle aches. She was recently started on an antihyperlipidemic agent. Which of the following is the likely lipid-lowering mechanism of her prescribed medication?

A. ● Increasin pression in the liver
B. ○ Inhibiting hormone-sensitive lipase
C. ○ Preventing cholesterol absorption
D. ○ Preventing intestinal absorption of bile acid
E. ○ Upregulating lipoprotein lipase

Hide Explanation

The correct answer is increasing cellular low-density lipoprotein (LDL) receptor expression in the liver (A), which describes the mechanism of action of statins. This drug inhibits 3-hydroxy-3-methylglutaryl-coenzyme A reductase and may cause myalgia as a side effect, especially in older patients. Inhibiting hormone-sensitive lipase (B) is niacin's mechanism of action, and it can cause myalgia when combined with statins but not alone. Preventing cholesterol absorption (C) is ezetimibe's mechanism of action, while preventing intestinal absorption of bile acid (D) is the mechanism of action of bile acid resins. Neither ezetimibe nor bile acid resins cause myalgia as a side effect. Upregulating lipoprotein lipase (E) is done by fibrates, which can also cause myopathy. However, fibrates may cause gallstones, making them an unsafe choice in this patient.

2. What is the primary mechanism by which statins reduce LDL levels?

A. ○ Decreased HDL levels
B. ● Inhibition
C. ○ Increased VLDL levels
D. ○ Inhibition of Acetyl-CoA-reductase

Hide Explanation

The correct answer is inhibition of HMG-CoA reductase (B). Statins inhibit the enzyme HMG-CoA reductase, which is the rate-limiting step in cholesterol synthesis, leading to decreased cholesterol production and increased LDL receptor expression. Decreased HDL levels (A) and increased VLDL levels (C) are incorrect. In fact, statins increase HDL levels and decrease circulating VLDLs. (D) Statins do no inhibit the function of Acetyl reductase block substrate binding for HMG-CoA reductase.

3. Which statin is metabolized by CYP3A4 ?

A. ○ Fluvastatin
B. ○ Pitavastatin
C. ○ Rosuvastatin
D. ● Simvastatin

Hide Explanation

The correct answer is Simvastatin (D). Fluvastatin, is primarily metabolized by CYP2C9. Pitavastatin (B) has minimal hepatic metabolism. Simvastatin (C), and Rosuvastatin (C) is primarily cleared by non-CYP450 transformations.

4. From the following, which statin has a lower SAMS risk

A. ○ Atorvastatin
B. ● Pravastatin
C. ○ Simvastatin
D. ○ Lovastatin

Hide Explanation

The correct answer is Pravastatin (B). Pravastatin is more hydrophillic. Atorvastatin (A), Simvastatin (C), and Lovastatin (D) have high SAMS risk since they are more lipophilic.

5. If a patient is taking Simvastatin to lower their cholesterol, it is best to advise them not to take large amounts of ___

A. ○ apple juice
B. ● grapefruit
C. ○ watermelon juice
D. ○ spinach

Hide Explanation

The correct answer is grapefruit juice (B). If ingested in large quantities, Grapefruit juice can inhibit CYP3A4. Since Simvastatin is metabolized by CYP3A4, the concurrent use of grapefruit juice can increase the adverse effects of simvastatin. The other options do not interfere with Simvastatin metabolism.

How would you rate this Brick?

References

- Bakillah A. *New Lipid Lowering Therapies For Cardiovascular and Metabolic Diseases.* MDPI; 2019.
- https://www.jacc.org/doi/10.1016/j.jacc.2022.07.006
- https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8931779/
- https://www.uptod tein-cholesterol-lowering-with-drugs-other-than-statins-and-pcsk9-inhibitors?search=hyperlipidemia%20medications&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1
- https://www.uptodate.com/contents/statins-actions-side-effects-and-administration?search=hyperlipidemia%20medications&source=search_result&selectedTitle=3~150&usage_type=default&display_rank=3

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the mechanism of action of statins in the context of managing hyperlipidemia and the rationale behind their use in preventing cardiovascular diseases.~~
02 ☑ ~~Classify statins based on their dose and discuss their Pharmacokinetic/pharmacodynamic profiles.~~
03 ☑ ~~Explain the adverse effects, contraindications, and precautions associated with statins and discuss strategies to minimize/prevent such effects.~~
04 ☑ ~~Explain the common reasons for statin intolerance and provide alternative therapeutic strategies.~~
05 ☑ ~~Identify the potential drug-drug interactions of statins and explain their underlying mechanisms.~~

electedTitle=1~150&usage_type=default&display_rank=1
- https://www.uptodate.com/contents/statins-actions-side-effects-and-administration?search=hyperlipidemia%20medications&source=search_result&selectedTitle=3~150&usage_type=default&display_rank=3

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the mechanism of action of statins in the context of managing hyperlipidemia and the rationale behind their use in preventing cardiovascular diseases.~~
02 ☑ ~~Classify statins based on their dose and discuss their Pharmacokinetic/pharmacodynamic profiles.~~
03 ☑ ~~Explain the adverse effects, contraindications, and precautions associated with statins and discuss strategies to minimize/prevent such effects.~~
04 ☑ ~~Explain the common reasons for statin intolerance and provide alternative therapeutic strategies.~~
05 ☑ ~~Identify the potential drug-drug interactions of statins and explain their underlying mechanisms.~~

Objective Confidence: 5/5